

Overview: Technical design specification detailing software architecture, database schema design, UX/UI frameworks, security blueprints, and integration pipelines for the LegalEase platform.

1. EXECUTIVE DESIGN OVERVIEW

The Design Phase defines the structural architecture, component interactions, database schemas, and interface workflows required to implement LegalEase's functional specifications safely, scalably, and securely.

1. HIGH-LEVEL SYSTEM ARCHITECTURE

- **Frontend Layer:** Single Page Application (SPA) built with React/TypeScript and Tailwind CSS, optimized for rapid document rendering and real-time collaborative editing.
- **API Gateway & Microservices:** REST & GraphQL API Gateway routing requests to modular backend microservices (Auth Service, Document Engine, AI Processing Worker, Workflow Coordinator).
- **AI & NLP Orchestration:** Asynchronous queue worker architecture using Celery/Redis connecting to LLM providers and vector databases (Pinecone/Milvus) for RAG semantic search.
- **Storage Infrastructure:** Multi-region PostgreSQL relational database for structured metadata and S3-compatible object storage with server-side encryption for document binary blobs.

2. DATABASE SCHEMA & MULTI-TENANCY ARCHITECTURE

- **Multi-Tenancy Model:** Schema-per-tenant separation in PostgreSQL ensuring isolation between law firm accounts and preventing cross-tenant data leaks.
- **Core Entities:** Users, Tenants, Matters/Cases, Document Templates, Generated Contracts, Audit Logs, Signature Requests.
- **Audit Logging Strategy:** Immutable event store capturing user actions, document views, edits, e-signatures, and AI query invocations for compliance audits.

3. USER EXPERIENCE (UX) & INTERFACE DESIGN BLUEPRINT

- **Workspace Dashboard:** Unified dashboard displaying active matters, pending contract reviews, unsigned e-signature requests, and recent activity streams.
- **Document Studio Interface:** Side-by-side view with document viewer on the left and AI Redlining/Clause Recommendations panel on the right.
- **Accessibility & Visual System:** High-contrast color palette, clean typography, WCAG 2.1 AA compliance, and keyboard-first shortcut navigation for power legal users.

4. INTEGRATION ARCHITECTURE & SECURITY DESIGN

- **E-Signature Webhooks:** Event-driven integration with DocuSign / Adobe Sign webhooks for status updates and auto-fetching completed PDFs.
- **Security Blueprint:** OAuth 2.0 + OIDC authentication, MFA enforcement, short-lived JWT tokens, and automated key rotation via AWS KMS / HashiCorp Vault.